

Tyler Rolfe

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EDUCATION

PhD in Machine Learning, University of Nottingham Expected 2030
School of Physics and Astronomy

- Probabilistic modelling for time series using non-Markovian methods such as state space models and large language models.
- Rare event analysis in foundation models using Monte Carlo methods and reinforcement learning

MSc Machine Learning in Science, University of Nottingham 2025
School of Physics and Astronomy (Distinction, 2nd in cohort)

- Strong foundation in linear algebra, probability, and statistical methods for machine learning.
- Deep learning architectures including CNNs, GANs, LSTMs, and transfer learning methods.
- Reinforcement learning techniques including tabular and neural network based approaches.
- Distributed machine learning using PySpark.

BSc Physics with Theoretical Astrophysics, University of Nottingham 2024
School of Physics and Astronomy (2:1)

- Computational physics techniques including numerical methods and simulations.
- Mathematical modelling of physical systems in areas including quantum physics, statistical mechanics, and astrophysics.
- Presentation of research via technical reports and presentations.

PROJECTS

Rare trajectory sampling via reinforcement learning

(*PhD*)

- Solved a stochastic excursion problem for a discreet random walker using reinforcement learning.
- Implemented policy iteration, REINFORCE, and actor-critic methods.
- Applied KL regularisation to control policy entropy and prevent collapse to deterministic policies.
- Validated results against exact solutions where possible.

Statistical Evaluation of AI Safety Techniques

(*Research Internship*)

- Implemented prompt-level red teaming strategies from HarmBench, including PAIR and TAP.
- Evaluated attack success rates across models under varying temperature settings.
- Assessed robustness of HarmBench methodologies, including LLM-as-judge evaluation schemes.

LLMs for constrained document generation with verification

(*Research Internship*)

- Developed an end-to-end pipeline for constrained document generation with automated verification.
- Stochastically generated executable target functions to ensure diversity in outputs.
- Used prompt engineering techniques to teach small language models how to structure human readable HTML tables

Transfer Learning for self-driving vehicles

(*MSc Machine Learning*)

- Applied transfer learning from large pre-trained image classifiers to a small driving dataset.
- Designed a dual-head architecture to control vehicle speed and steering.
- Deployed model to Pi-Car hardware, accounting for real-time constraints such as latency.

Work Experience

Business Analyst Budgeting Solutions Ltd 2025–2026

- Contributed to financial planning and budgeting processes for a range of clients.
- Maintained and updated complex financial models.
- Managed multiple concurrent projects under time constraints, ensuring clear communication with stakeholders.

Research Intern University of Nottingham 2025

- Awarded competitive internship position alongside MSc due to strong academic performance and technical interview.
- Conducted research on large language models through both independent and collaborative projects.
- Contributed to broader research discussions and development of group research directions.

Part-Time Work Experience, Mid Wales & Nottingham 2018–2024

- Customer-facing roles including barista and bartender positions in fast-paced environments.
- Developed strong interpersonal, teamwork, and conflict resolution skills.
- Managed responsibilities alongside academic commitments, demonstrating effective time management and reliability.

VOLUNTEERING

Physics Mentor, University of Nottingham 2023–2024

- Provided academic and pastoral support to five first-year students.
- Guided students on course material and study approaches.

Vice President, Welsh Society, University of Nottingham 2023–2024

- Supported coordination of society activities and oversight of committee members.
- Contributed to organisation of events and delivery of society initiatives.
- Assisted in overseeing projects and maintaining effective operations.

Social Secretary, Welsh Society, University of Nottingham 2022–2023

- Led recruitment through in-person and online channels, increasing membership by 40%.
- Organised social events and coordinated collaborations with other societies.

IMS Football Player/Captain, University of Nottingham 2021–2025

- Captained an intramural football team, organising fixtures and players, often at short notice.
- Also represented the IMS Physics team, playing a key part in reaching two finals.